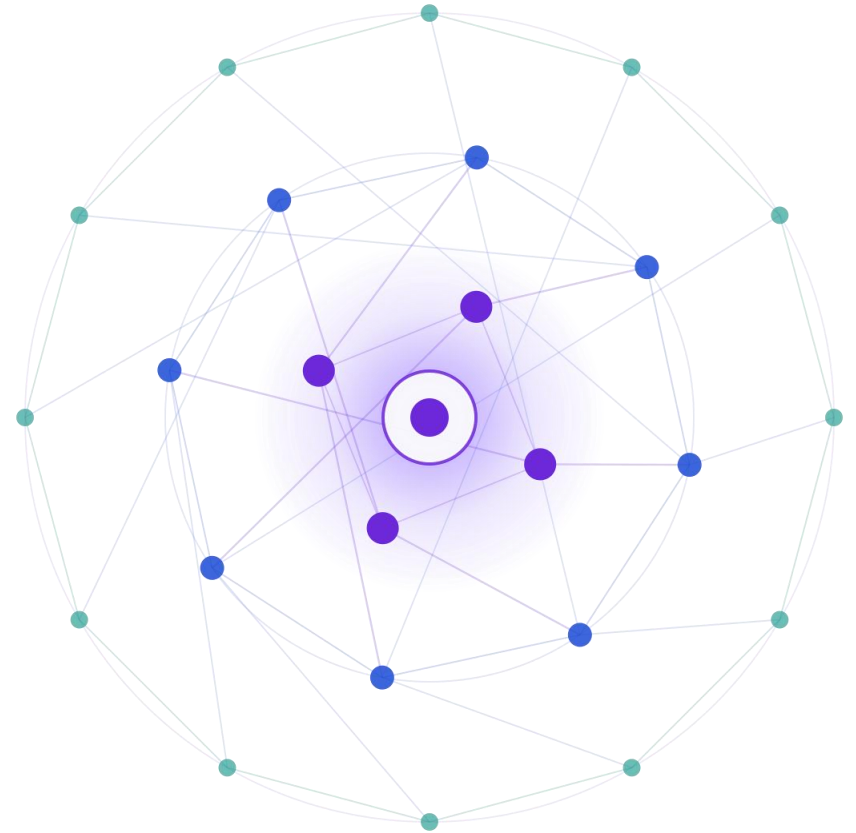


Generative AI Hackathon

3rd Edition 2026 – 27

- **Project - Keel:** voice assistants that act only when you've finished
- **Theme ID - Theme 05:** Interruptible Real-Time Agents
- **Team Name** - Hackatoons
- **College Name** – VIT Vellore
- **Member Name & Email 1-** Harshit Aryan (harshitaryan21@gmail.com)
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- **Member Name & Email 3-** Aashman Biyani (biyaniaashman@gmail.com)
- **Member Name & Email 4-** Hamsika Gokavarapu (hamsika.g7@gmail.com)
- **Submission Github link** - github.com/PROSTLE/samsung



Theme



Illustration, AI-generated

“Looking at flights to Miami on October 5th. Oh, wait. My schedule just changed... so actually make it October 7th instead.”

A real caller from the benchmark’s recordings. Many assistants run the October 5th search before the caller has finished, and a search or booking that has already gone out can’t be taken back.

People don’t talk in clean sentences.

We pause, say “um”, start again and change our minds halfway through. An assistant that acts on the first thing it hears books the wrong day, searches the wrong city, or does the same thing twice.

Theme 05 asks for three things



Respond quickly

spoken feedback within a few hundred milliseconds, no dead air



Keep talking while it works

tools run in the background, never blocking the conversation



Recover cleanly

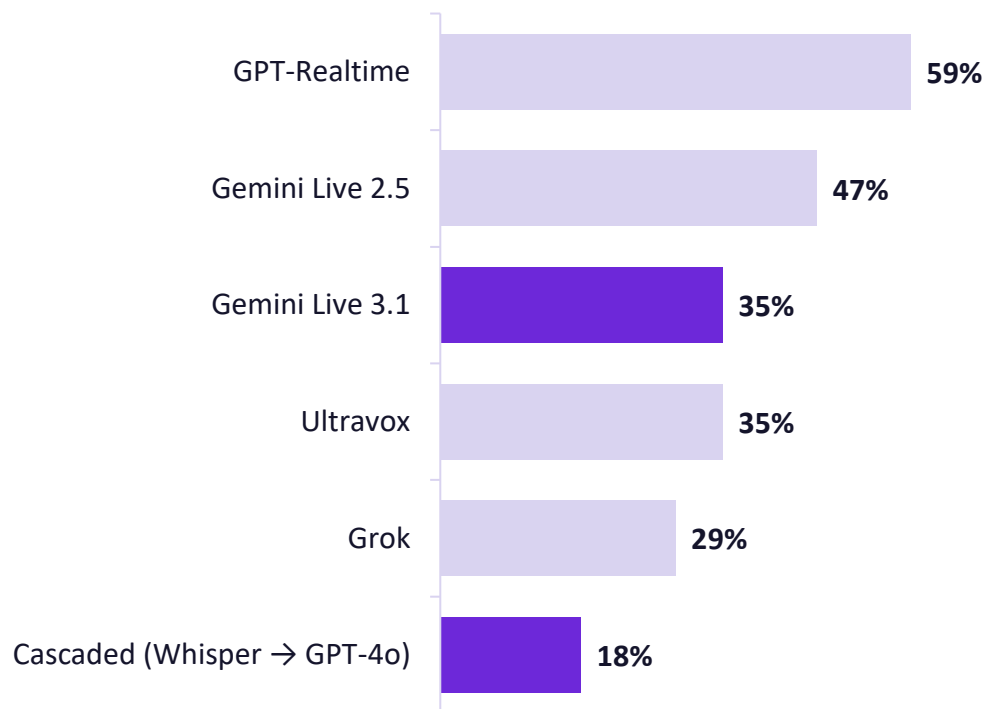
drop what the user took back, and never repeat an action

Our answer: think fast, but act only once you’ve finished.

Existing Solutions & Gaps

When people correct themselves, assistants fail

Pass rate on the self-correction recordings, as published in the benchmark paper



Highlighted: the two systems our set-ups build on. Source: Full-Duplex-Bench v3 paper, Table 3.



Speech-to-text commits too early

In the classic pipeline the transcript is final before the correction arrives, so the AI never hears it.



Realtime models act before you finish

The paper caught Gemini Live 3.1 calling a tool 2.27 s before the user had finished correcting, locking in the wrong city.



Frameworks leave it to every developer

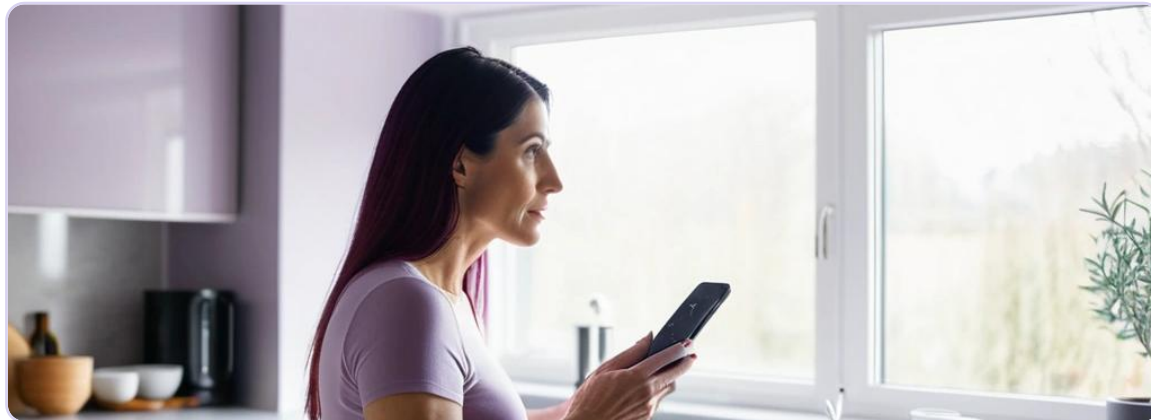
Voice frameworks either cancel on any interruption or keep going. Each team has to add safety rules tool by tool.

Even the best system passes only 60% of the whole benchmark.

Our Solutions & Architecture Diagram

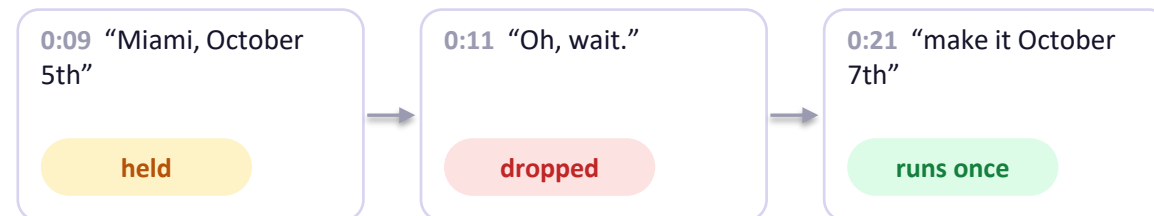
Keel: think fast, act carefully

Keel sits between the assistant's brain (the language model) and its hands (the tools). The model can plan as early as it likes; Keel decides when an action is allowed to happen.



Illustration, AI-generated

A real recording, step by step



Problem

Gaps

Solution

How it works

Use case

Results

Next



Waits until you're done

An action the AI plans while you're still talking is held until you finish.



Drops what you take back

Keep talking or correct yourself, and the held action is thrown away. It never runs.



Does it only once

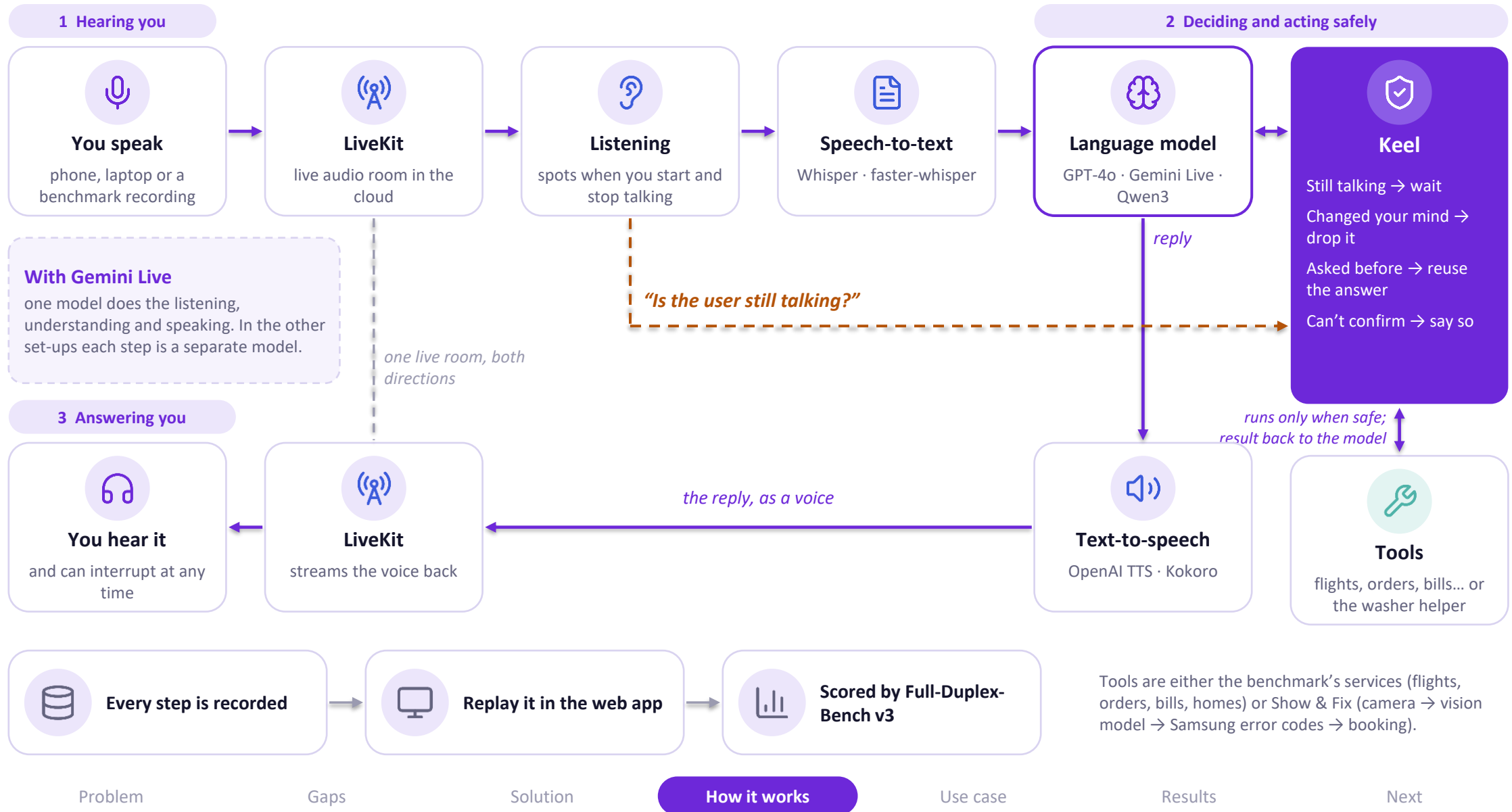
Asking twice never books twice: the first result is reused.



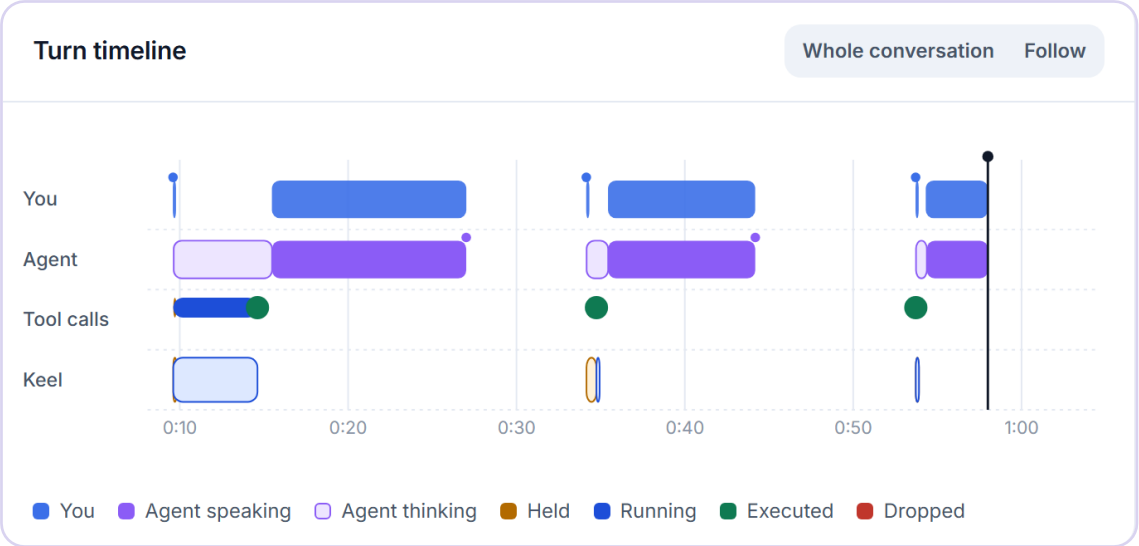
Never fakes success

If it can't confirm an action worked, it says so instead of saying "done".

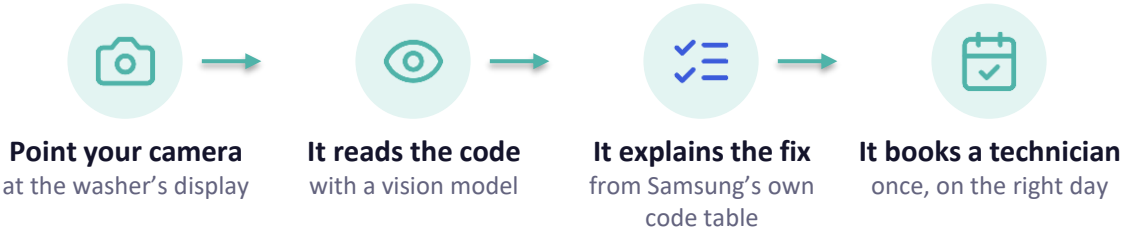
How it works, end to end



Impact & Use case



The same live session on Keel's timeline: the display is read, then the booking is held until the user finishes "Friday... oh, wait, Saturday", then made once.



Show & Fix

Our extension: a washer helper that can see

Transcript

You 0:34

Thank you. Please book a technician for Friday morning. Oh, wait. No, make it Saturday morning.

Agent 0:44

All set! A technician is booked for Saturday morning. Your booking ID is SVC-0F7C85. Anything else I can help with?

You 0:53

Did the booking go through?



Test display used as the camera (labelled synthetic)

- ✓ Read the code as 5E, though the display showed "SE"
- ✓ Booked Saturday once after "Friday... oh, wait, Saturday"
- ✓ Checked before answering "did it go through?"

A live session, recorded end to end on Gemini Live

Where else this matters

- In-car: change the destination mid-sentence
- Support lines: change a booking without double-booking

Innovation highlights, results and limitations

48/100

strict pass, measured
answers must match word for word

66/100

compared by meaning
our estimate; the official run uses a judge

100%

turns answered
same model in the paper: 78%

5%

talks over the user
same model in the paper: 19%

89%

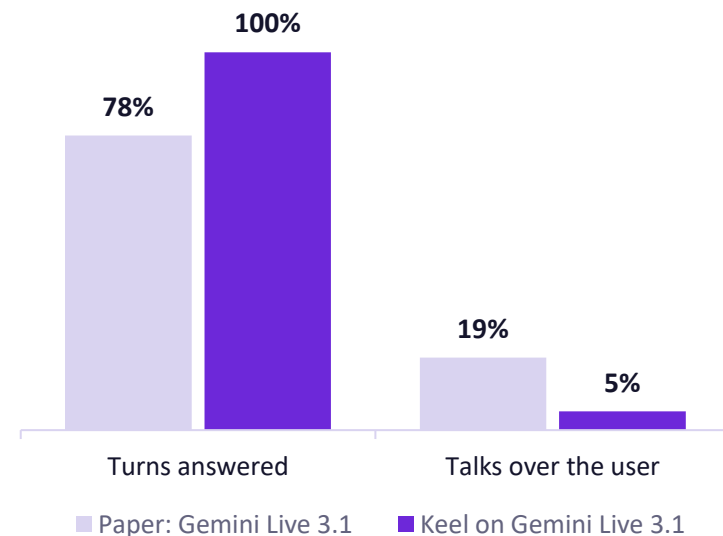
right tool chosen
on answered turns

4.2 s

first reply, median
fix in progress: an instant “one moment”

Full run of all 100 benchmark recordings on Gemini Live 3.1 after this week's fixes (run 20260928T211335). Our first full run, before them, scored 43/100 strict.

Same model, same test

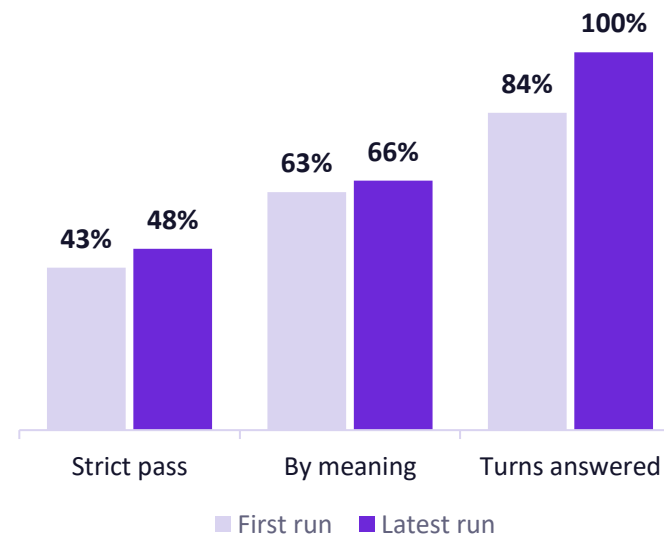


Problem

Gaps

Solution

Before and after our fixes



How it works

Use case

What's new

Acts on the final intent

actions wait for the user, and taken-back ones never run

Knows what “oh, wait” means

it waits for the correction that follows

Works with any model

cascaded, realtime or fully local

Where we're still weak →

Results

Next

What's next

What we're still working on



Speed The first reply comes after 5.1 s on average; the paper's Gemini Live 3.1 manages 3.95 s. Keel's own wait is 0.6 s of it (median).

We're working on: Say a short "one moment" straight away, and find out why the end of a turn is reported late.



Right answers, written differently 18 of our misses are correct answers in another format, like "2026-07-15" for "July 15", the format the tool itself asks for.

What to expect: The official judge compares by meaning and accepts other date formats; counting those gives 66/100.



Silent turns Every conversation got a reply in our latest run (first run: 16 silent; the paper: 22 on the same model).

What we did: In our first run 9 were our network: the agent joined after the caller had finished. We stopped the background uploads competing with the audio and let the agent take calls while the machine is busy. Since then: no late joins.



Self-corrections Now 53% with word-for-word matching (first run: 24%).

We're working on: Measuring how long to wait after "oh, wait" on fresh recordings, never on the test set.



Illustration, AI-generated

Coming up



The organisers' official, judged run



A full run on local models, on a proper GPU



A real washer and a real camera



The 3–5 minute demo video

Problem

Gaps

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